

Programme: Diploma in Cloud Computing and Big Data	
Course Code: 6271B	Course Title: Artificial Neural Networks
Semester: 6	Credits: 4
Course Category: Programme elective course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart Artificial Neural Network (ANN) Concepts
- To provide concepts on Deep Learning Architectures and applications

Course Prerequisites:

Topic	Course Code	Course name	Semester
Calculus and coordinate geometry	1002	Mathematics I	1
Problem Analysis and Programming concepts	2131	Problem Solving and Programming	2
Linear Algebra	2002	Mathematics II	2
Programming in Python	3279	Programming in Python Lab	3
Concept of data, data structures and algorithms	4272	Data Structures and Algorithms	4
Big Data and data analytics	4271	Big Data Concepts	4
Implementation of data structures	4278	Data Structures and Algorithms Lab	4
Artificial Intelligence (AI) Concepts		Machine Learning	5
Practical knowledge in implementing AI techniques		Machine Learning Lab	5

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate Artificial Neuron inspired from Biological Neuron	14	Understanding
CO2	Describe Basics of Artificial Neural Network	13	Understanding
CO3	Demonstrate Convolutional Neural Network	16	Applying
CO4	Outline Recurrent Neural Network	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate Artificial Neuron inspired from Biological Neuron		
M1.01	Compare Structure of Artificial Neuron with Biological Neuron	3	Understanding
M1.02	Discuss Perceptron and Multi Level Perceptron (MLP)	3	Understanding

M1.03	Discuss Feedforward network and Backpropagation	4	Understanding
M1.04	Discuss Loss function and Activation function	4	Understanding
Contents: Biological neurons, Logical computations with neurons- McCulloch and Pitts Neuron, Artificial Neuron, Perceptron - Linear Threshold Unit (LTU)- Perceptron Learning Rule, MLP- Multilayer Feedforward network, Backpropagation, Learning algorithms -Gradient Descent, Loss function/cost function - Mean Squared Error (MSE), Entropy, Cross Entropy Activation functions - Step function, Sigmoid, Tanh, ReLU, Leaky ReLU, Softmax			
CO2	Describe concepts of Artificial Neural Network.		
M2.01	Outline Deep Neural Network (DNN) Models	4	Understanding
M2.02	Describe design issues of backpropagation learning	5	Understanding
M2.03	List Applications of DNN	4	Remembering
	Series Test – I	1	
Contents : Models of Artificial Neural Networks- DNN, Convolutional Neural network (CNN), Recurrent Neural Network (RNN), Design issues- Challenges - Vanishing Gradient and Exploding Gradient Applications of Deep Neural Network- Language modeling, Speech recognition, Visual tracking, Video capturing, Image capturing, Image processing, Object detection, Object recognition and Medical Diagnosis			
CO3	Demonstrate the concepts in Convolutional Neural Network		
M3.01	Illustrate Architecture of Convolutional Neural Network	4	Understanding
M3.02	Use Convolution Operation using different Filters	4	Applying
M3.03	Demonstrate various types of Pooling	3	Applying
M3.04	Describe Applications, Advantages and Disadvantages of CNN	2	Applying
M3.05	Compare various CNN pre-trained models	3	Understanding

Contents: Convolutional Neural Network Architecture- Input layer, Hidden layers, Convolution layer, Pooling layer, Output layer; Striding and Padding. Types of Filters, Convolution operation, Pooling operation Applications of CNN- Image classification, Advantages - feature Extraction , multi-class image classification. Disadvantages- requirement of large dataset Pre-trained models- ResNet, VGG, GoogleNet, Xception.			
CO4	Outline Recurrent Neural Network (RNN)		
M4.01	Illustrate Architecture of RNN	3	Understanding
M4.02	Elaborate Selective Read, Write and Forget	3	Understanding
M4.03	Explain how does Long Short Term Memory (LSTM) solve the issue like Vanishing Gradient	3	Understanding
M4.04	Describe working of Autoencoder	3	Understanding
M4.05	Summarize Applications of RNN, LSTM and Autoencoder	3	Understanding
	Series Test – II	1	
Contents : Recurrent Neural Network (RNN) concepts, memory cells, Selective Read, Write, Forget. Concept and working of LSTM, Auto Encoder Applications of RNN- Natural language Processing, Image capturing, Video classification Application of Autoencoder - Clustering, Feature extraction			

Text / Reference

T/R	Book Title/Author
R1	Deep Learning by Ian Goodfellow, Yoshua Bengio and Aaron Courville, The MIT Press
R2	Machine Learning by Tom Mitchell, McGraw-Hill Press, 1997